

12-WAGL-4400-003

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GENERAL NOTES (FOR CONTINUATION, SEE SHEET 2)

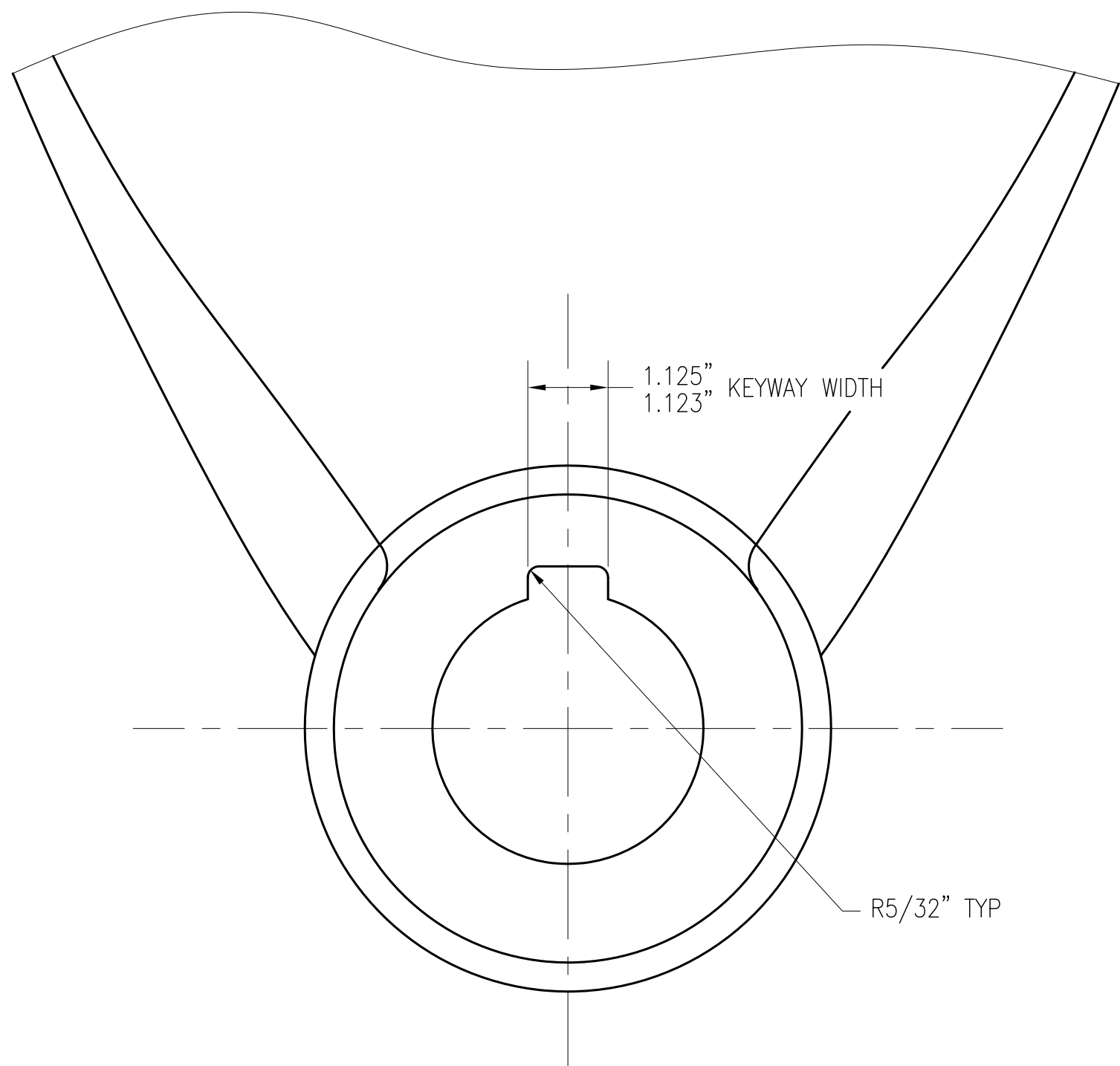
- PROPELLERS ARE PROCURED TO ELC SPECIFICATION P-245-0146
- A SHIP'S SET OF PROPELLERS SHALL BE CONSIDERED AS ONE (1) LEFT HANDED (LH) AND ONE (1) RIGHT HANDED (RH).
- PROPELLER IS DESIGNED FOR 675 HP AT 2100 RPM ENGINE WITH 4:1 RATIO REVERSE REDUCTION GEAR
- NATIONAL STOCK NUMBERS (NSN) ARE:
RIGHT HAND: 2010-01-534-9089
LEFT HAND: 2010-01-534-9095

DATA

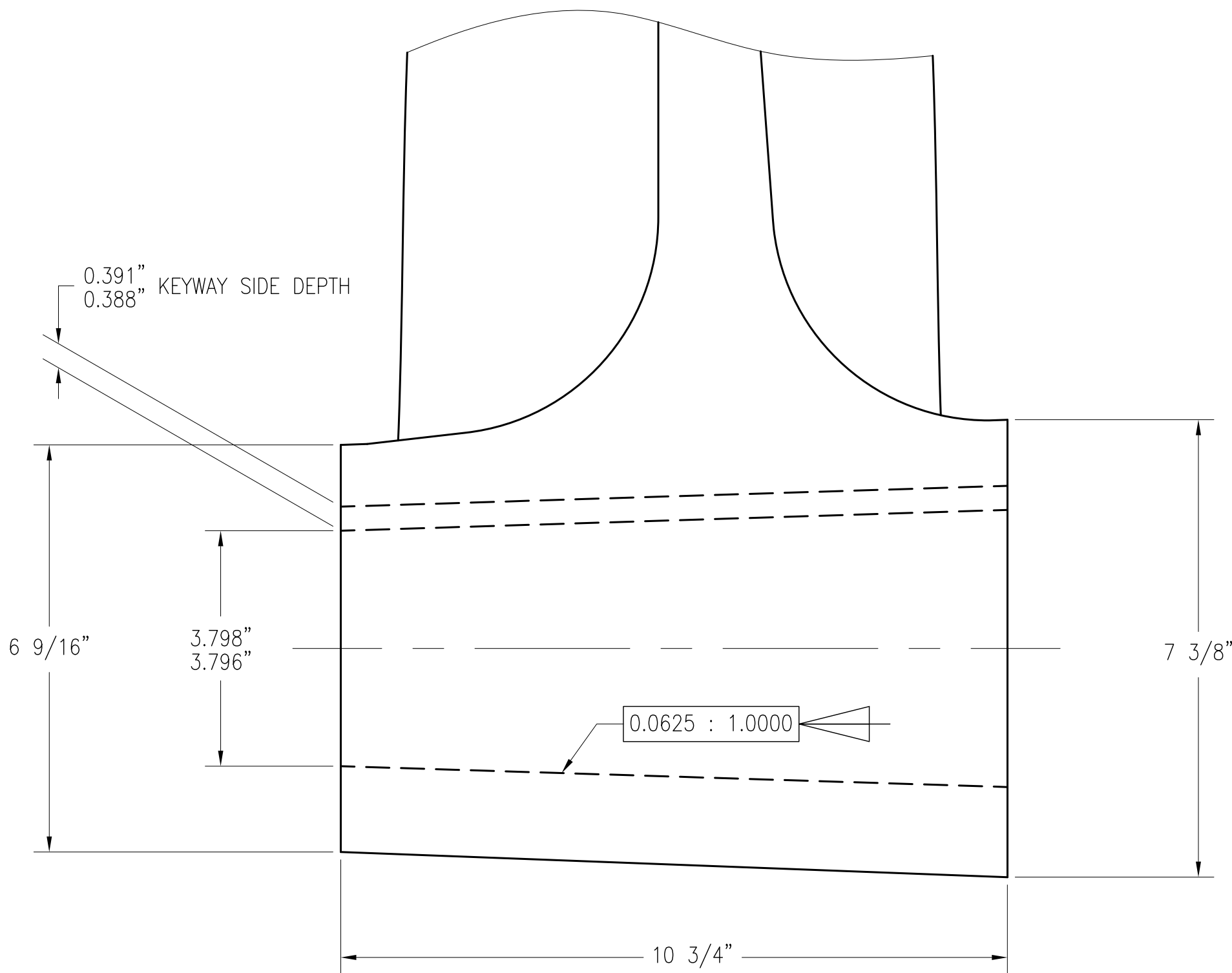
DIAMETER 48"
PITCH 40"
NO. BLADES 4
ROTATION RH & LH
D.A.R. 0.71 (MINIMUM)
PROP. MAT'L NI-AL-BRZ (ABS GR.4)
FAIRWATER MAT'L NI-AL-BRZ (ABS GR.4)

TOLERANCES IF NOT OTHERWISE NOTED

FRACTIONAL: $\pm 1/64"$



SECTION 7-A
PROPELLER
SCALE: 6" = 1'-0"



ELEVATION 4-A
PROPELLER
SCALE: 6" = 1'-0"

THIS DRAWING SUPERSEDES
USCG DRAWINGS
75-WAGL-4400-1 AND
75-WAGL-4400-2

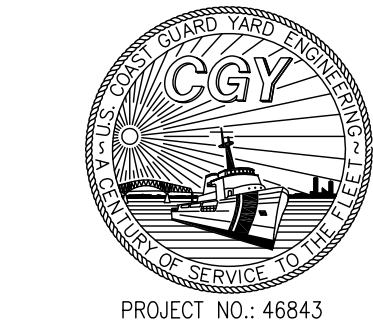
1 2 3 4 L L

1 2 3 4 L L

1 2 3 4 L L

SHT	REV	STATUS
4	L	
3	L	
2	L	
1	L	

65B WD	65505 OSAGE	65506 SANGAMON
65 WD	65503 OBION	65504 SCIOTO
75E WLR	75408 PATOKA	75409 CHENA
	75406 KICKAPOO	75407 KANAWHA
75D WAGL	75306 CLAMP	75309 HATCHET
		75307 WEDGE
75C WAGL	75405 CHEYENNE	75404 CHIPPEWA
	75403 WYAONDA	75401 GASCONADE
		75402 MUSKINGUM
75B WAGL	75305 VISE	
75-WAGL	75303 SLEDGE	75304 MALLET
75-WLIC	75301 ANVIL	75302 HAMMER
75-WLIC	75310 AXE	
PLAN-SET		HULLS
APPLICABILITY		



DESIGNED:	
DRAWN: JOC	5/16/06
CHECKED:	
APPROVED:	
U.S.C.G. APPROVAL	8/22/06
WALT WUJEK	
CHIEF - YARD MACHINERY BRANCH	

UNITED STATES COAST GUARD		WASHINGTON, D.C. 20593	
OFFICE OF NAVAL ENGINEERING			
75		WLIC/WLR	
PROPELLER TO SUIT IF 1312 MDE			
SIZE	FSCM CAGE NO	U.S.C.G. DRAWING NO.	REV
D	81340	75-WAGL-4400-003	L
SCALE: 6" = 1'-0"		6.00 SQ FT	SHEET 1 OF 4

REVISIONS					
REV	SHT	ZONE	ITEM	DESCRIPTION	DATE
A	1	4-A	1	EXTENDED APPLICABILITY TO VISE, AXE, MUSKINGUM.	12/11/07
B	1	4-A	1	EXTENDED APPLICABILITY TO HATCHET, CLAMP & MALLET.	4/11/08
C	1	4-A	1	EXTENDED APPLICABILITY TO HAMMER & PATOKA.	10/3/08
D	1	4-A	1	EXTENDED APPLICABILITY TO CHIPPEWA, OBION & SANGAMON.	4/15/09
E	1	4-A	1	EXTENDED APPLICABILITY TO OUACHITA & SLEDGE AND KICKAPOO.	9/17/09
F	1	4-A	1	EXTENDED APPLICABILITY TO SCIOTO, CIMARRON & CHENA.	2/24/10
G	1	4-A	1	EXTENDED APPLICABILITY TO KANAWHA & OSAGE.	8/17/10
H	1	4-A	1	EXTENDED APPLICABILITY TO CHEYENNE & WEDGE.	12/13/10
J	1	4-A	1	EXTENDED APPLICABILITY TO WYAONDA & GASCONADE.	5/13/11
K	1	6-A	1	DUE TO COMPLETION OF IRESS UPDATES, THIS DWG SUPERSEDES 75-WAGL-4400-1 & 75-WAGL-4400-2.	9/19/14
L	1	4-A	1	IRESS DRAWINGS WAS INCORPORATED TO DRAWING.	03/27/15
				REVISED STATUS AND ADDED NEW SHEETS 2, 3, & 4.	
				ADDED NOTE TO INDICATE CONTINUATION TO SHT 2.	
				ADDED REFERENCES.	

NO	TITLE	DRAWING NO
3	PROPELLER DETAILS	75-WLIC_245_1
2	PROP SHAFT DETS (75401)	75C-WAGL_4301_2
1	PROP SHAFT & STERN TUBE A&D	75E-WLR_4301_1

REFERENCE PLANS

SPECIAL NOTATIONS

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12-WAGL-4400-003

3-WAGL-4400-003
REV L

GENERAL NOTES

5. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
6. THE PROPELLER BLADES SHALL BE CAST FROM NICKEL ALUMINUM BRONZE, TYPE 4, IN ACCORDANCE WITH ABS "RULES FOR BUILDING AND CLASSING STEEL VESSELS".
7. THE PROPELLER BLADE SURFACE ROUGHNESS SHALL NOT EXCEED 63 MICRO-INCHES Ra. PROPELLER HUB EXTERIOR SURFACE SHALL NOT EXCEED 125 MICRO-INCHES Ra.
8. BLADE ROOT FILLETS ARE TO BE MODIFIED GRADUALLY AROUND THE BLADE SECTION AND SHALL BE FAIR. THE FILLET RADIUS SHALL NOT BE LESS THAN 1-1/4 INCH.
9. THE SURFACE FINISH OF THE KEY WAY AND SHAFT CONTACT AREA OF THE BORE SHALL NOT EXCEED 63 MICRO-INCHES (Ra)

IDENTIFICATION MARKINGS

10. THE FOLLOWING DATA SHALL BE STAMPED, IN CHARACTERS 6 TO 12mm HIGH, ON THE HUB OUTSIDE DIAMETER NEAR THE AFT END, CLEAR OF THE FILLETS:
- a. SERIAL NUMBER
 - b. SHIP CLASS
 - c. STARBOARD OR PORT
 - d. DIRECTION OF ROTATION (RH OR LH) WHEN VIEW AFT LOOKING FWD.
 - e. NATIONAL STOCK NUMBER (PROVIDED BY APPLICABLE SPEC)
 - f. DESIGN DIAMETER - 48 INCHES
 - g. DESIGN PITCH AT 0.7R - 40 INCHES
 - h. NAME OF MANUFACTURER AND DATE OF MANUFACTURE
 - i. ACTUAL WEIGHT IN POUNDS
 - j. BLADE MATERIAL (NIBRAL ABS TYPE 4)

TOLERANCES

11. PITCH SHALL NOT VARY MORE THAN PLUS OR MINUS 1.5 PERCENT AT EACH INDIVIDUAL BLADE RADIUS.
12. PITCH VARIATION BETWEEN ADJACENT DESIGN SECTIONS SHALL NOT EXCEED 0.75 PERCENT.
13. INDIVIDUAL BLADE MEAN PITCH SHALL NOT VARY MORE THAN PLUS OR MINUS 1.0 PERCENT OF DESIGN PITCH.
14. MEAN PITCH VARIATION ON PROPELLER BLADE SET SHALL NOT EXCEED 0.5 PERCENT.
15. THE MAXIMUM BLADE THICKNESS AT EACH DESIGN RADIUS SHALL BE WITHIN +2% / -1% , WITH A MINIMUM OF +0.08 INCH / -0.04 INCH.
16. BLADE SECTION WIDTH SHALL BE ACCURATE WITHIN +/- 1.5% WITH A MIN. OF +/- 0.25 INCH.
17. THE RADIUS SHALL BE ACCURATE WITHIN A TOLERANCE OF +0 / -1.0 PERCENT OF DESIGN.
18. THE ANGULAR SPACING BETWEEN THE CENTERS OF THE BLADES SHALL BE 90 DEGREES, PLUS OR MINUS 0.5 DEGREE.
19. EACH BLADE SHALL TRACK WITHIN 1/2 INCH AT THE TIP.
20. EDGE AND TIP GAGE CLEARANCE SHALL NOT EXCEED 0.32 INCH..
21. CLEARANCE UNDER THE FAIRING ROD SHALL NOT EXCEED 0.063 INCH AND SHALL NOT VARY AT A RATE GREATER THAN 0.01 INCH PER INCH.
22. TOLERANCES ARE NOT CUMULATIVE. WHERE TWO OR MORE TOLERANCES CONFLICT THE CLOSER TOLERANCE SHALL GOVERN. UNLESS OTHERWISE SPECIFIED THE TOLERANCE ON ALL DIMENSIONS SHALL BE AS FOLLOWS:
- ANGULAR TOLERANCES: +/- 15 SEC
- DIMENSIONAL TOLERANCES: 2-PLACE +/- 0.01 INCH
3-PLACE +/- 0.006 INCH
- FRACTIONAL TOLERANCES: +/- 0.02 INCH

GENERAL NOTES CONT'D

BALANCE

23. MATERIAL REMOVED TO EFFECT BALANCE SHALL BE TAKEN FROM THE SUCTION FACE ON AS LARGE AN AREA AS PRACTICABLE AND SHALL NOT APPROACH THE LEADING OR TRAILING EDGES WITHIN (10) TEN PERCENT OF THE BLADE WIDTH.
24. CARE SHALL BE TAKEN TO ASSURE THAT REMOVAL OF MATERIAL SHALL NOT CAUSE THE TOLERANCES FOR THICKNESS, SURFACE CONTOUR, EDGE CONTOUR, WIDTH OR PITCH TO BE EXCEEDED.
25. THERE SHALL BE NO MACHINING OF THE PROPELLER AFTER FINAL BALANCE CHECK.
26. THE PROPELLER BLADES SHALL BE SINGLE-PLANE BALANCED, IMBALANCE SHALL NOT EXCEED 6.11 oz-in..

INSPECTION

27. MEASUREMENTS SHALL BE TAKEN ON EACH FINISHED BLADE AT EACH DESIGN RADIUS AND RECORDED TO VERIFY COMPLIANCE WITH THE DRAWING DIMENSIONS AND TOLERANCES. DESIGN RADII SHALL BE INTERPRETED AS THOSE RADII LISTED IN THE TABLE OF OFFSETS.
28. DIMENSIONAL INSPECTION OF THE PROPELLER BLADES SHALL BE ACCOMPLISHED AFTER ANY REQUIRED BALANCE CORRECTIONS HAVE BEEN ACCOMPLISHED, IF BALANCING IS ACCOMPLISHED AFTER DIMENSIONAL INSPECTION, ONLY THE AREAS AFFECTED BY THE BALANCING MUST BE REINSPECTED DIMENSIONALLY TO ASSURE CONFORMANCE WITH THE DESIGN.

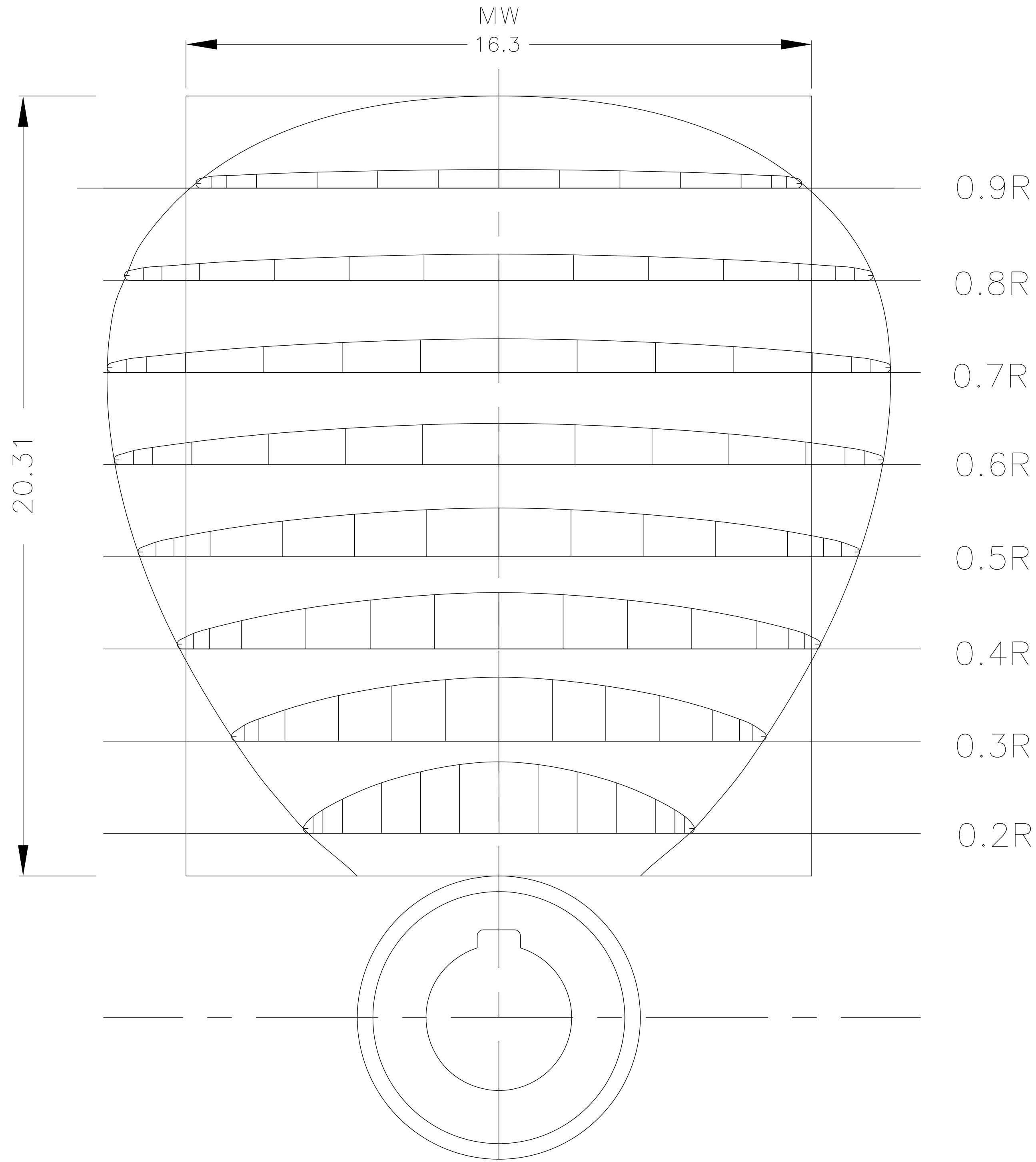
INSTALLATION

29. DAMAGE TO THE PROPELLER BLADE EDGES AND TIPS SHALL BE AVOIDED AT ALL TIMES.
30. THE PRESERVATIVE COATING AND EDGE PROTECTORS SHALL BE LEFT ON DURING INSTALLATION AND REMAIN UNDISTURBED UNTIL THE VESSEL IS READY TO BE REFLOATED.
31. PRIOR TO FINAL INSTALLATION, THE PROPELLER TO SHAFT FIT SHALL BE CHECKED WITH KEY INSTALLED USING A THIN LAYER OF PRUSSIAN BLUE. A UNIFORMLY DISTRIBUTED CONTACT IMPRESSION OF AT LEAST 80% IS REQUIRED ON BOTH THE FORWARD AND AFT AREAS. THE PROPELLER MAY BE ADVANCED UP TO 0.04 in, IF NECESSARY, TO ACHIEVE THE REQUIRED CONTACT IMPRESSION. AT FINAL ASSEMBLY THE PROPELLER BORE AND SHAFT TAPER SURFACES SHALL BE CLEAN, DRY AND FREE OF LUBRICANTS. THE PROPELLER SHALL BE ADVANCED BY MEANS OF A HYDRAULIC DEVICE SUCH AS THE MAPECO PILGRIM NUT. THE REQUIRED ADVANCE MEASURED WITH A DIAL INDICATOR SHALL BE:

ADVANCE (in) = $D_s/160 [1-.0052 (T_f-60)]$
Ds= SHAFT DIA. (in)
Tf= AMBIENT TEMP, DEG. F
OR ADVANCE (in) = (Ds)(K)
WHERE
K= COEFFICIENT FROM TABLE

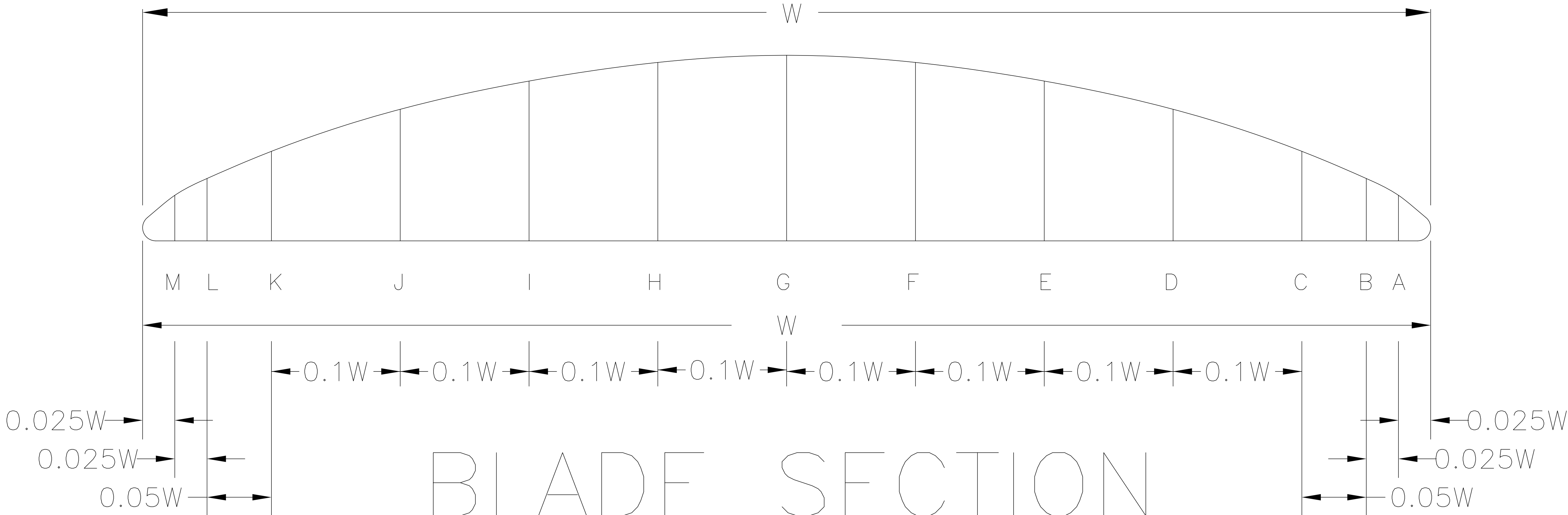
PROPELLER ADVANCE COEFFICIENTS

AMBIENT TEMPERATURE °F	K
20.0	0.00756
30.0	0.00723
40.0	0.00691
50.0	0.00658
60.0	0.00625
70.0	0.00593
80.0	0.00561
90.0	0.00527



FOR ABS CLASSIFICATION
AT 0.35 r/R: Cs=.7441
Cn=.1030
ABS CERTIFICATE REQUIRED : ICE CLASS B0
MAXIMUM ENGINE OUTPUT, 100% MCR: 675 BHP at 2100 RPM

INERTIA (WR ²) IN AIR	1719	lb-ft ²
INERTIA (WR ²) IN WATER	2242	lb-ft ²
NUMBER OF BLADES	4	
DIAMETER - D	48	in
NOMINAL PITCH P0.7	40	in
P/D 0.7R	0.833	
RPM	525	
EXPANDED AREA - Fa	9.328	ft ²
WEIGHT OF PROPELLER INCLUDING HUB	412-430	lbs
BLADE MATERIAL	NiAlBr ABS TYPE 4	
TENSILE STRENGTH	85,000 (Min)	* lb/in ²
EXPANDED BLADE AREA RATIO	0.73	
MEAN WIDTH RATIO	0.36	

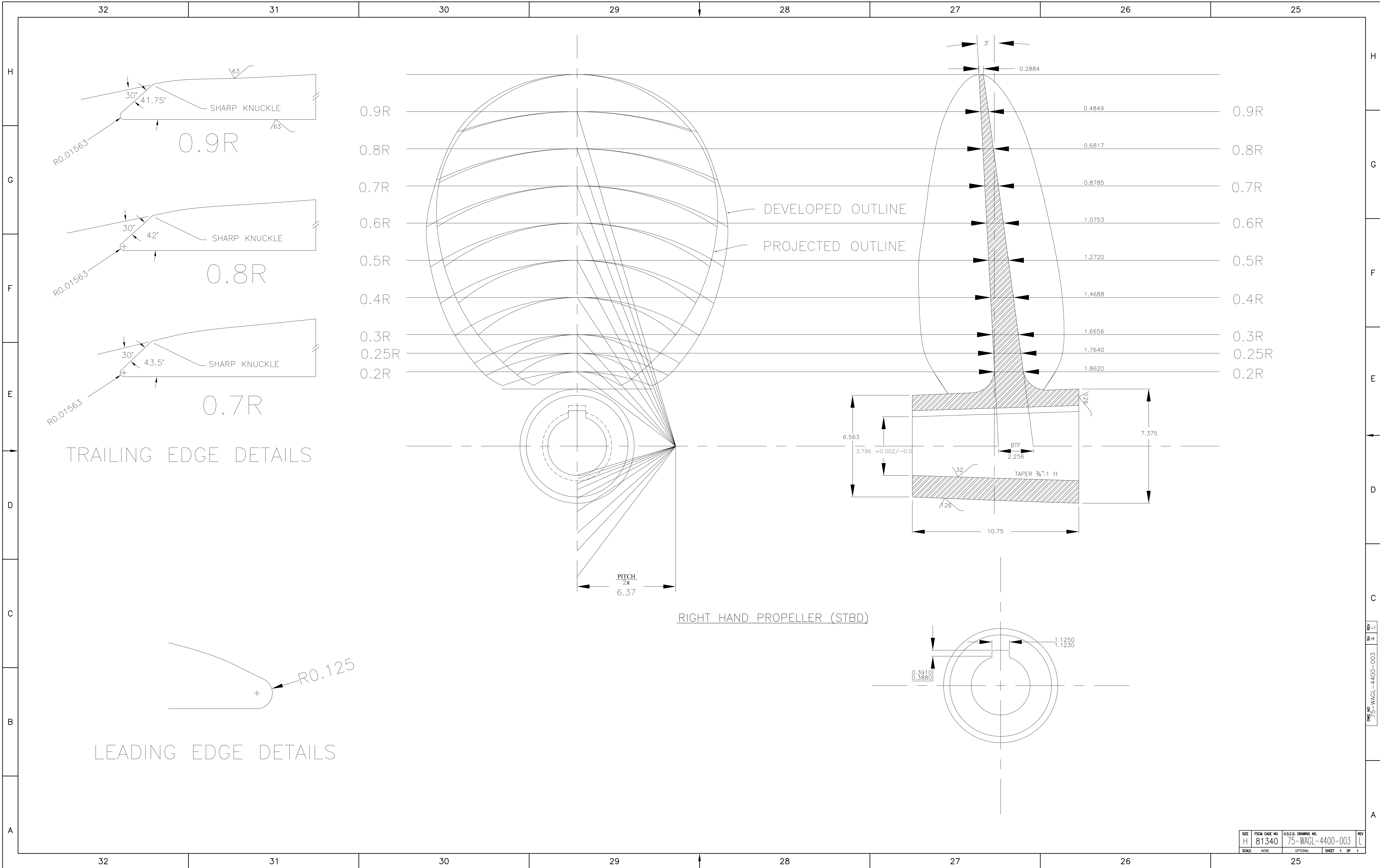


r/R		TER	TE	M	L	K	J	I	H	G	F	E	D	C	B	A	LE	LER
0.9	Width	1/64	15.782	15.387	14.993	14.204	12.626	11.047	9.469	7.891	6.313	4.735	3.156	1.578	0.789	0.395	0.000	1/8
	Thickness		0.000	0.3073	0.3287	0.3631	0.4164	0.452	0.476	0.485	0.476	0.452	0.4164	0.3631	0.3287	0.3073	0.000	
0.8	Width	1/64	19.503	19.015	18.528	17.553	15.602	13.652	11.702	9.752	7.801	5.851	3.901	1.95	0.975	0.488	0.000	1/8
	Thickness		0.000	0.3267	0.3693	0.4381	0.5448	0.6163	0.664	0.682	0.664	0.6163	0.5448	0.4381	0.3693	0.3267	0.000	
0.7	Width	1/64	20.406	19.896	19.386	18.365	16.325	14.284	12.244	10.203	8.162	6.122	4.081	2.041	1.020	0.510	0.000	1/8
	Thickness		0.000	0.346	0.410	0.5132	0.6732	0.7804	0.851	0.878	0.851	0.7804	0.8016	0.5132	0.410	0.346	0.000	
0.6	Width	1/32	20.040	19.539	19.028	18.036	16.032	14.028	12.024	10.020	8.016	6.012	3.760	2.004	1.002	0.501	0.000	1/8
	Thickness		0.000	0.3653	0.4507	0.5883	0.8016	0.945	1.039	1.075	1.039	0.945	0.930	0.5883	0.4507	0.3653	0.000	
0.5	Width	1/8	18.801	18.331	17.861	16.921	15.041	13.161	11.281	9.401	7.520	5.640	3.351	1.880	0.940	0.47	0.000	1/8
	Thickness		0.000	0.3847	0.4913	0.6633	0.930	1.1097	1.227	1.272	1.227	1.1087	1.0584	0.6633	0.4913	0.3847	0.000	
0.4	Width	1/8	16.755	16.336	15.917	15.080	13.404	11.729	10.053	8.378	6.702	5.027	3.351	1.676	0.838	0.419	0.000	1/8
	Thickness		0.000	0.404	0.532	0.7384	1.0584	1.273	1.414	1.469	1.414	1.273	1.0584	0.7384	0.532	0.404	0.000	
0.3	Width	1/8	13.931	13.583	13.234	12.538	11.145	9.752	8.3586	6.966	5.5724	4.1793	2.786	1.393	0.6966	0.3483	0.000	1/8
	Thickness		0.000	0.4233	0.5727	0.8135	1.187	1.437	1.602	1.666	1.602	1.437	1.187	0.8135	0.5727	0.4233	0.000	
0.25	Width	1/8	12.241	11.935	11.629	11.017	9.793	8.569	7.345	6.121	4.896	3.672	2.448	1.224	0.612	0.306	0.000	1/8
	Thickness		0.000	0.433	0.593	0.851	1.251	1.519	1.696	1.764	1.696	1.519	1.251	0.851	0.593	0.433	0.000	
0.2	Width	1/8	10.188	9.934	9.679	9.169	8.151	7.132	6.113	5.094	4.075	3.057	2.038	1.019	0.509	0.255	0.000	1/8
	Thickness		0.000	0.443	0.613	0.889	1.315	1.601	1.79	1.862	1.79	1.601	1.315	0.889	0.613	0.443	0.000	

TABLE OF OFFSETS

SIZE	FSM CASE NO	U.S.C.G. DRAWING NO.	REV
H	81340	75-WAGL-4400-003	L
SCALE	NONE	OPTIONAL	SHEET 3 OF 4

\\P-WAGL-4400-003
DWG NO 4
VSR



SIZE	FSM CAGE NO	U.S.C.G. DRAWING NO.	REV
H	81340	75-WAGL-4400-003	L
SCALE	NONE	OPTIONAL	SHEET 4 OF 4